CHAPTER 5

TABULATION AND DIAGRAMMATIC REPRESENTATION

CHAPTER ESSENCE AT A GLANCE:
Master formal statistical table anatomy, one-dimensional and two-dimensional diagrams (Bar, Pie), and frequency graphs (Histogram, Polygon, Ogives for Median).

4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Components of a Statistical Table:** A formal table consists of Table Number, Title, Captions (column headings), Stubs (row headings), Body (numerical data cells), Headnote, and Footnote with Source note.
- **Bar Diagrams (One-Dimensional):** Data represented through bars of equal width where height or length is proportional to magnitude; includes Simple, Multiple, Subdivided (Component), and Percentage bar diagrams.
- **Pie Diagram (Two-Dimensional Area):** A circular diagram divided into sectors where each sector angle is proportional to component percentage (Angle = [Component Value divided by Total] * 360 degrees).
- **Frequency Graphs: Histogram and Ogives:** Histograms represent continuous distributions with adjacent rectangles whose areas reflect frequency; Ogives plot cumulative frequencies, intersecting at the Median.

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$\text{Pie Sector Angle} = \frac{\text{Component Value}}{\text{Total Value}} \times 360^\circ$$

Two-Dimensional Circular Angular Conversion Formula

$$\text{Percentage Bar Component} = \frac{\text{Sub-Component Value}}{\text{Total Magnitude}} \times 100$$

Standardized 100 Percent Relative Component Height

$$\text{Ogive Intersection Point} \iff \text{Less-Than Ogive} \cap \text{More-Than Ogive} = (\text{Median}, \tfrac{N}{2})$$

Graphical Median Determination Property

$$\text{Histogram Adjusted Frequency} = \frac{\text{Class Frequency}}{\text{Class Width}} \times \text{Standard Width}$$

Frequency Density Adjustment for Unequal Class Widths

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** In histograms, rectangles must be contiguous without spaces between them because the data is continuous; bar diagrams have distinct gaps between bars.
- **Alert:** Pie diagram sector angles must always sum to exactly 360 degrees; always check arithmetic totals before drawing.
- **Alert:** The intersection of 'Less-than' and 'More-than' ogives on the horizontal X-axis gives the exact value of the Median.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Describe the main parts of a formal statistical table. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **1. Table Number and Title:** Clear identifier and prominent heading stating what, where, and when the data refers to. **[1 Mark]**

• **2. Captions and Stubs:** Caption designates column headings and units; Stub designates row designations on the extreme left. **[1 Mark]**

• **3. Body:** The main interior part of the table containing numerical cell entries organized into rows and columns. **[1 Mark]**

• **4. Headnote, Footnote, and Source:** Headnote provides unit specifications; Footnote explains specific figures; Source note states original collecting agency. **[1 Mark]**

2. Distinguish between Subdivided (Component) Bar Diagram and Multiple Bar Diagram with suitable use cases. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Subdivided Bar Diagram:** Each bar represents total magnitude of a phenomenon, partitioned into colored or shaded segments proportional to components (e.g. Total household budget broken down into Food, Rent, Education). **[1.5 Marks]**

• **Multiple Bar Diagram:** Two or more separate adjacent bars are drawn for each category to show comparison between interrelated variables (e.g. Comparing Import and Export values across consecutive years). **[1.5 Marks]**

3. The monthly expenditure of a family is: Food = Rs 4,000, Rent = Rs 2,500, Clothing = Rs 1,500, Education = Rs 2,000. Calculate the central angles for constructing a Pie Diagram. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Total Expenditure:** 4,000 + 2,500 + 1,500 + 2,000 = Rs 10,000. **[0.5 Mark]**

• **Formula:** Angle = (Component Value divided by Total) * 360 degrees. **[0.5 Mark]**

• **Angle Calculations:**
• Food: (4,000 divided by 10,000) * 360 = 144°.
• Rent: (2,500 divided by 10,000) * 360 = 90°. **[1 Mark]**

• • Clothing: (1,500 divided by 10,000) * 360 = 54°.
• Education: (2,000 divided by 10,000) * 360 = 72°. Total = 144 + 90 + 54 + 72 = 360°. **[1 Mark]**

4. Distinguish between a Histogram and a Bar Diagram. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Data Nature:** Histograms represent continuous frequency distributions; Bar diagrams represent discrete variables or categorical attributes. **[1 Mark]**

• **Spacing:** Histograms feature contiguous rectangles without gaps; Bar diagrams have uniform gaps between adjacent bars. **[1 Mark]**

• **Dimensionality:** Histogram is a two-dimensional graph where area of rectangle represents frequency; Bar diagram is one-dimensional where only height represents magnitude. **[1 Mark]**

5. Explain how the Median of a frequency distribution is determined graphically using Ogives. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Method 1 (Two Ogives):** Construct both the 'Less-than' ogive and 'More-than' ogive on the same graph sheet with marks on the X-axis and cumulative frequency on the Y-axis. **[1.5 Marks]**

• **Intersection Point:** From the intersection point of the two curves, drop a perpendicular to the horizontal X-axis. The point where the perpendicular meets the X-axis gives the exact value of the Median. **[1.5 Marks]**

6. Can Mode be determined graphically? Explain briefly how it is located using a Histogram. **[Score: 2 Marks]**

Answer & Key Valuation Steps:

• **Procedure:** Yes, Mode is located by identifying the tallest rectangle (modal class) in a histogram. **[1 Mark]**

• **Intersection:** Join the top right corner of the modal rectangle to the top right corner of the preceding rectangle, and join the top left corner of the modal rectangle to the top left corner of the succeeding rectangle. Drop a perpendicular from their intersection to the X-axis to obtain the Mode. **[1 Mark]**